

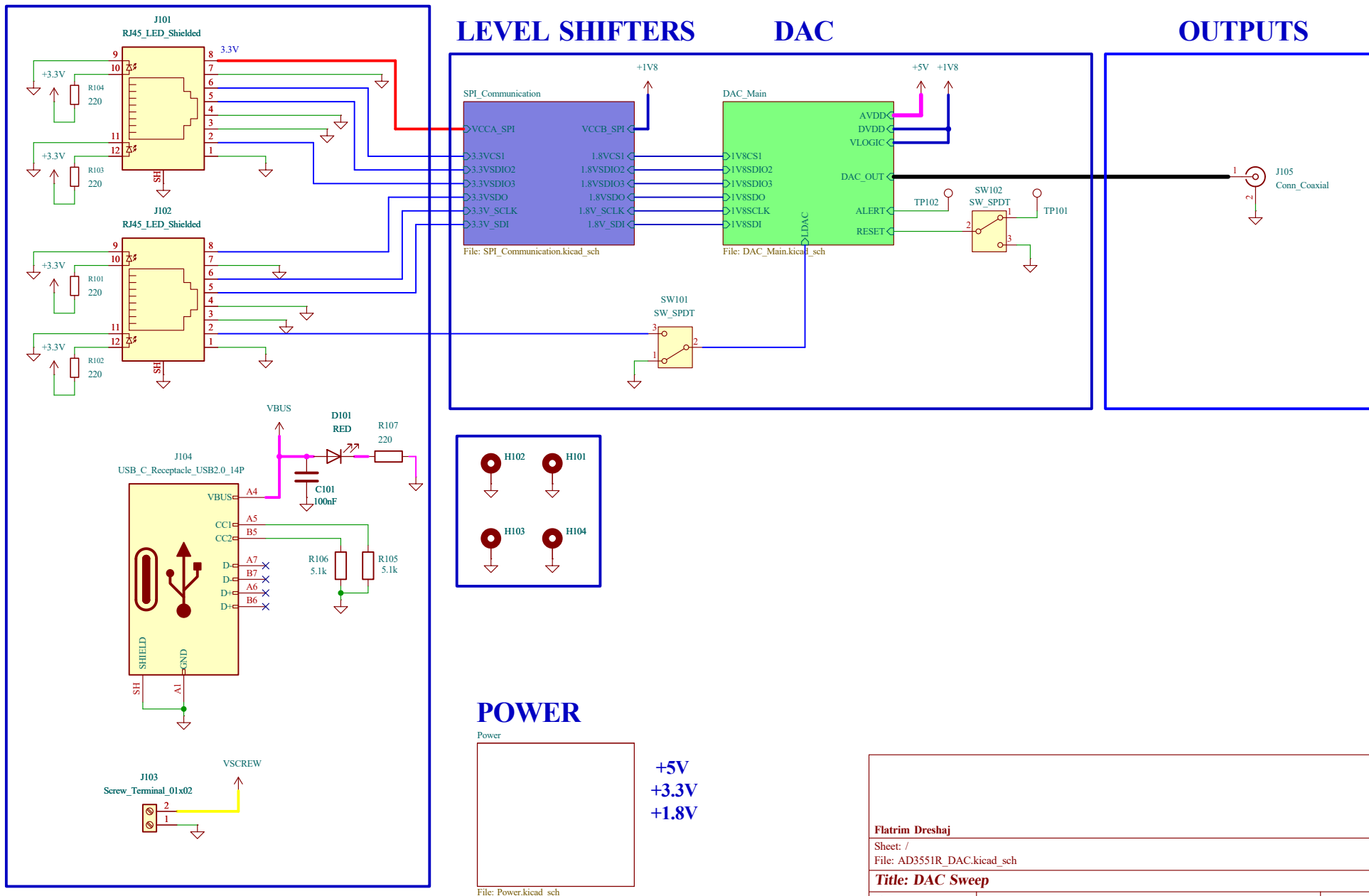
INPUTS

AD3551R DAC Fast Sweep Board

LEVEL SHIFTERS

DAC

OUTPUTS



Flatrim Dreshaj

Sheet: /
File: AD3551R_DAC.kicad_sch

Title: DAC Sweep

Size: A4
KiCad E.D.A. 10.0.5

Date: 2026-08-22

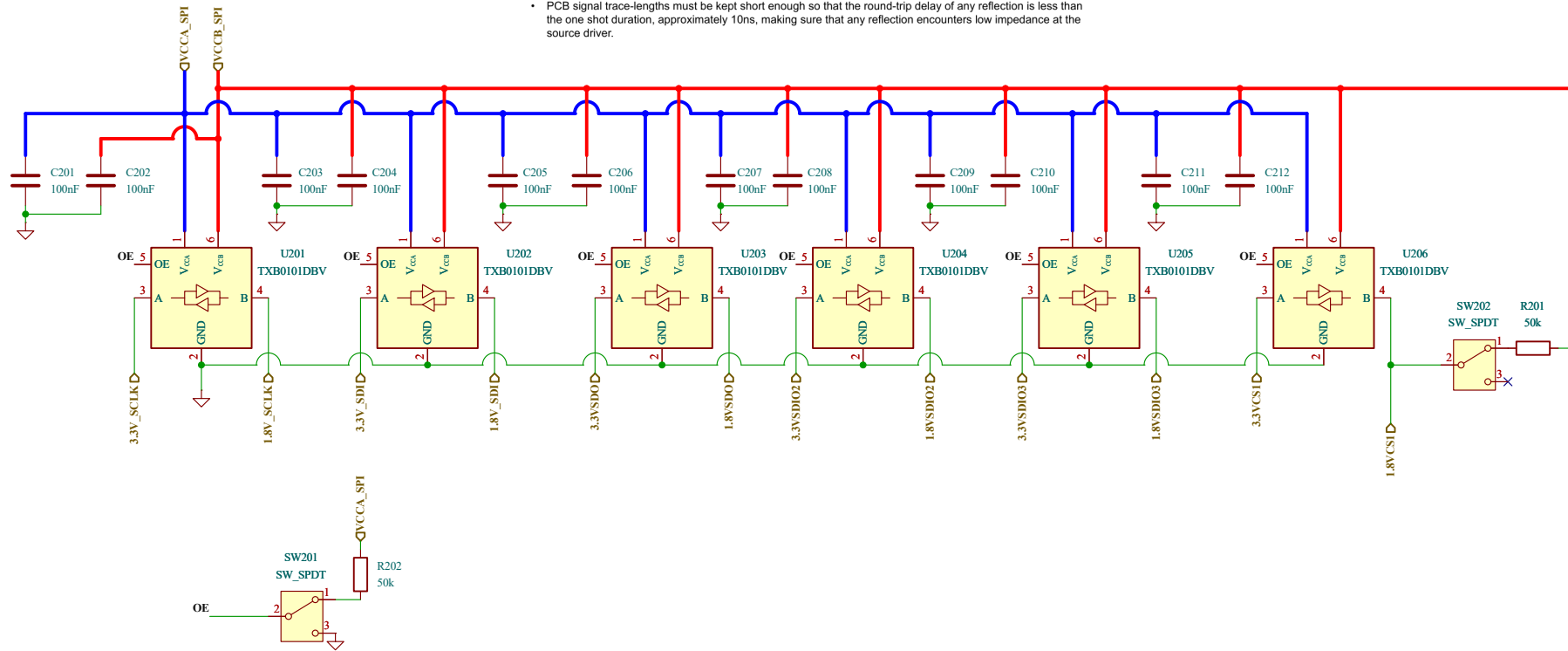
Rev: 0

Id: 1/4

7.4.1 Layout Guidelines

For device reliability, the following common printed-circuit board layout guidelines are recommended.

- Bypass capacitors should be used on power supplies, and should be placed as close as possible to the V_{CCA} , V_{CCB} pin and GND pin.
- Short trace lengths should be used to avoid excessive loading.
- PCB signal trace-lengths must be kept short enough so that the round-trip delay of any reflection is less than the one shot duration, approximately 10ns, making sure that any reflection encounters low impedance at the source driver.



Sheet: /SPI_Communication/
File: SPI_Communication.kicad_sch

Title:

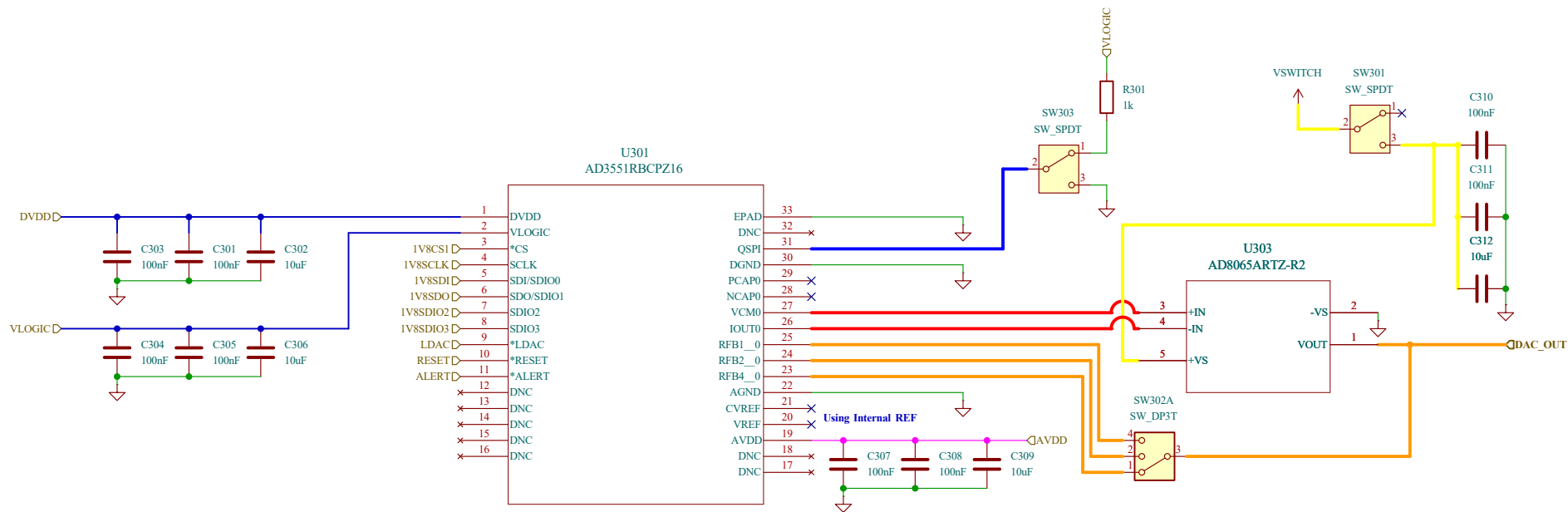
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Rev:

Id: 2/4



Layout Notes:

QFN Package Extend pads

I_Out as short and thin as possible due to amplifier slew.

Route Line with constant characteristic impedance for timing

Sheet: /DAC_Main/
File: DAC_Main.kicad_sch

Title:

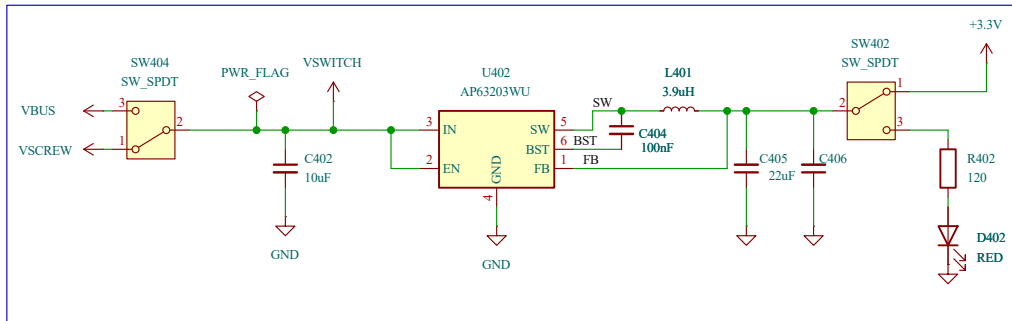
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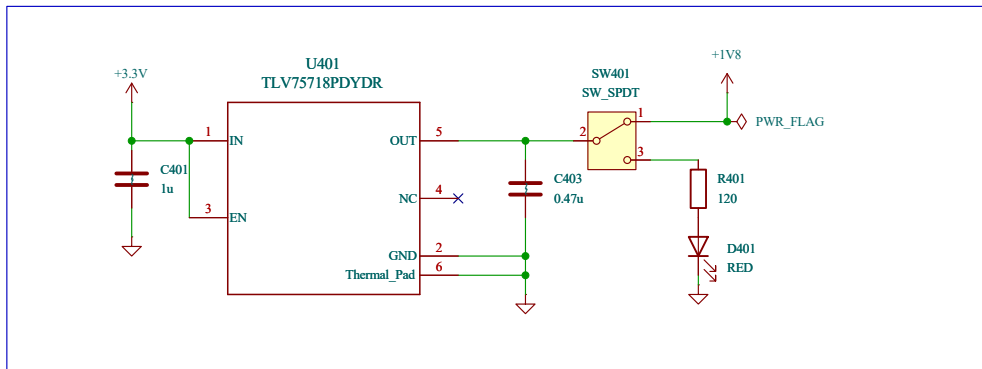
KiCad E.D.A. 10.0.5

Rev:

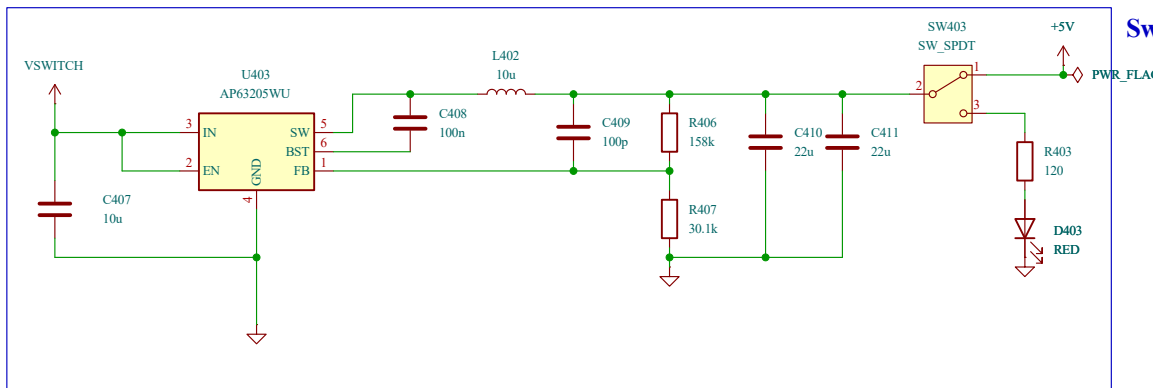
Id: 3/4



Buck: 3.3V 2A



LDO +1.8V 1A



Switching: 5V 2A

Total power Consumption

DAC:
LEVEL SHIFTERS:
LEDs:

TOTAL

5V:
3.3V:
1.8V:

Sheet: /Power/
File: Power.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 10.0.5

Rev:

Id: 4/4